AI-Driven Wind Turbine Failure Prediction using Neural Networks

ReneWind and Deep Learning

April 19, 2026

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Executive Summary

Business Goal

- The objective is to build a neural network model to **predict wind turbine generator failures** using sensor data
- Multiple neural network architectures were built and compared across different configurations
- Models were progressively improved by adding **depth, dropout regularization, class weighting, and optimizer changes**

Key Insights

- The dataset is **heavily imbalanced** — only ~5.5% of cases are actual failures
- **Recall** was chosen as the evaluation metric to minimize missed failures
- **Adam optimizer** outperformed SGD in terms of recall and generalization

Final Outcome

- **Model 4** (Adam optimiser, 2 hidden layers, no dropout) was selected as the best-performing model

Business Recommendations

- Deploy Model 4 for real-time turbine monitoring to catch failures before they occur
- Prioritize early inspection when the model flags a potential failure to avoid costly replacements
- Improve calibration of sensors V1 and V2 which had missing values
- Periodically retrain the model as new sensor data accumulates

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Business Problem Overview and Solution Approach

Problem Definition

- **The wind turbine monitoring process aims to:**
  - ✓ Predict generator failures before they occur
  - ✓ Reduce costly unplanned replacements
  - ✓ Minimise downtime through proactive maintenance
  - ✓ Support data-driven maintenance scheduling
- **This is a binary classification problem:**
  - ✓ **Target variable:** Failure Status (Failure = 1, No Failure = 0)
- **Cost structure:**
  - ✓ True Positive → Repair cost (medium)
  - ✓ False Negative → Full replacement (highest cost)
  - ✓ False Positive → Inspection only (lowest cost)

Solution Approach / Methodology

- Data Understanding & Overview
- Exploratory Data Analysis (EDA)
- Data Cleaning & Preprocessing
- Missing Value Imputation
- Train-Validation Split (stratified 80/20)
- Model Building — 7 Neural Network architectures
- Progressive improvement: depth → dropout → class weights → optimiser change
- Model Evaluation using Recall metric
- Final Model Selection based on validation performance

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EDA Results – Univariate Analysis

Target Variable: Failure Status

- The dataset shows a strong class imbalance:
  - ✓ Majority class: ~94.5% No Failure (Class 0)
  - ✓ Minority class: ~5.5% Failure (Class 1)
- Similar distribution in test set — 94.36% vs 5.64%
- Confirms the need for Recall as the primary metric rather than Accuracy

Numerical Features (V1–V40)

- All 40 features are continuous and numerical (float64)
- Features are sensor readings — encrypted/ciphered for confidentiality
- Several features exhibit:
  - ✓ Skewed distributions
  - ✓ Outliers visible in boxplots
  - ✓ Wide variation in scale across features

Missing Values

- Missing values identified only in V1 and V2
- Present in both train and test datasets
- Treated using median imputation

Key Univariate Insights

- Strong class imbalance — accuracy alone would be misleading (~94% by predicting all zeros)
- Sensor features show varied distributions and outliers across all 40 variables
- Only 2 out of 40 features had missing values — data is largely complete

Link to – Appendix slide

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EDA Results – Bivariate Analysis

Correlation Analysis

- Correlation heatmap computed across all 40 sensor features
- Some feature pairs show moderate correlation
- No extreme multicollinearity observed
- Neural networks handle correlated features well through learned weights

Class Distribution vs Features

- Failure cases (Class 1) represent a small but distinct minority
- The failure pattern is not random — it is learnable from sensor readings
- Features contribute independently to the prediction

Key Bivariate Insights

- No single feature perfectly separates failure from non-failure
- A combination of all 40 features is needed — justifying a neural network approach
- The low failure rate (~5.5%) makes Recall the critical success metric over Accuracy

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Data Preprocessing

Duplicate Value Check

- Checked for duplicate rows in the training dataset
- No duplicates found. All 20,000 rows are unique observations

Missing Value Detection & Treatment

- Missing values identified in V1 and V2 only
- Treated **using median imputation,** robust to outliers
- **Imputer** fitted only on X_train
- Applied separately to X_val and X_test to prevent data leakage
- Verified no missing values remain before modeling

Outlier Detection & Treatment

- Outliers detected via boxplots across all V1–V40 features
- No explicit removal applied
- Neural networks are relatively robust to outliers
- Removing outliers risks losing rare failure signals from the minority class

Feature Engineering

- No additional feature engineering performed
- All 40 sensor features used as is
- No encoding needed. All features are already numerical

Data Preparation for Modeling

Data Splitting:

- Training Set — 16,000 rows (80%)
- Validation Set — 4,000 rows (20%)
- Test Set — separate 5,000 rows held out for final evaluation
- Stratified split applied (random_state=1) to preserve class ratios in each split
- Class Imbalance Handling:
- Class weights computed from training labels
- Used in Models 3 and 6 to bias the model toward minority class (failures)

Final Insight

- Preprocessing ensured clean, properly structured data with no leakage, enabling fair and reliable model comparison

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Model Evaluation Criterion

Common Settings Across All Models

- Epochs: 50
- Batch Size: 32
- Loss Function: **Binary Crossentropy**
- Output Layer: 1 neuron with **Sigmoid** activation
- Evaluation Metric: **Recall**

Model 0 — Baseline

- 1 Hidden Layer: 7 neurons, ReLU activation
- Optimizer: **SGD**
- Dropout: No
- Class Weight: No

Model 1 — Added Depth

- 2 Hidden Layers: 16 → 8 neurons, ReLU activation
- Optimizer: **SGD**
- Dropout: No
- Class Weight: No

Model 2 — Dropout Regularization

- 3 Hidden Layers: 16 → 8 → 4 neurons, ReLU activation
- Dropout: 50% after first hidden layer
- Optimizer: **SGD**
- Class Weight: No

Model 3 — Class Weighting

- 3 Hidden Layers: 16 → 8 → 4 neurons, ReLU activation
- Dropout: 50% after first hidden layer
- Optimizer: **SGD**
- Class Weight: Yes

Model 4 — Adam Optimiser (Best Model)

- 2 Hidden Layers: 16 → 8 neurons, ReLU activation
- Optimizer: **Adam**
- Dropout: No
- Class Weight: No

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Model Evaluation Criterion (cont.)

Model 5 — Deeper + Dropout + Adam

- 3 Hidden Layers: 32 → 16 → 8 neurons, ReLU activation
- Dropout: 50% after first hidden layer
- Optimizer: **Adam**
- Class Weight: No

Model 6 — Class Weight + Adam

- 3 Hidden Layers: 32 → 16 → 8 neurons, ReLU activation
- Dropout: 50% after first hidden layer
- Optimizer: **Adam**
- Class Weight: Yes
- Model 6 follows a similar architecture to Model 5 with the addition of class weighting
- The performance did not show a significant improvement over Model 4
- Adding class weights on top of an already well-performing Adam optimizer did not yield meaningful gains in this case
- Model 4 remained the stronger choice based on validation recall

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Model Performance Summary

Key Observations

- Models progressively improved as depth, regularization and optimizer changes were introduced
- Model 0 established a solid baseline with a single hidden layer
- Model 1 showed a clear recall improvement simply by adding a second hidden layer
- Model 2 introduced dropout which helped regularization but slightly reduced recall
- Model 3 recovered recall by incorporating class weights to handle the imbalance
- Model 4 achieved the highest validation recall by switching to the Adam optimizer
- Model 5 added more depth and dropout with Adam but did not surpass Model 4
- Model 6 introduced class weighting on top of Model 5 however the performance remained comparable with no significant improvement over Model 4

Training Set Performance

Model	Accuracy	Recall	Precision	F1 Score
Model 0	0.9865	0.8916	0.9764	0.9294
Model 1	0.9921	0.9354	0.9879	0.9600
Model 2	0.9868	0.8849	0.9876	0.9294
Model 3	0.9666	0.9341	0.8183	0.8656
Model 4	**0.9942**	**0.9497**	**0.9946**	**0.9710**
Model 5	0.9913	0.9239	0.9929	0.9554
Model 6	0.9913	0.9239	0.9929	0.9554

Validation Set Performance

Model	Accuracy	Recall	Precision	F1 Score
Model 0	0.9868	0.8976	0.9725	0.9314
Model 1	0.9885	0.9176	0.9698	0.9419
Model 2	0.9873	0.8936	0.9827	0.9331
Model 3	0.9633	0.9191	0.8064	0.8523
Model 4	**0.9913**	**0.9318**	**0.9835**	**0.9559**
Model 5	0.9893	0.9095	0.9866	0.9443
Model 6	0.9893	0.9095	0.9866	0.9443

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Best Model - Test Performance

Key Observations

- Model 0 — Simple baseline; decent recall but limited by shallow architecture
- Model 1 — Adding a second layer noticeably improved recall
- Model 2 — Dropout reduced overfitting but slightly lowered recall
- Model 3 — Class weights helped recover recall on imbalanced data
- Model 4 — Switching to Adam optimizer gave the highest validation recall
- Model 5 — Deeper with dropout; slightly lower recall than Model 4
- Model 6 — Excluded due to notebook bug

Final Model Selected: Model 4

- Architecture: Input (40) → Hidden Layer 1 (16, ReLU) → Hidden Layer 2 (8, ReLU) → Output (1, Sigmoid)
- Optimizer: Adam
- **Loss**: Binary Crossentropy
- Epochs: 50 | Batch Size: 32

Reason for Selection

- Highest validation recall of 93.18% — catches the most actual failures on unseen data
- Minimal overfitting — train-validation recall gap of only 1.8%
- Best balance between recall, precision, and generalization
- Adam optimizer converges faster and more effectively than SGD for this problem

Final Insight

- **Model 4** generalizes well — test recall (93.17%) closely matches validation recall (93.18%), confirming no overfitting
- The model successfully identifies 93 out of every 100 actual turbine failures
- Directly reduces costly unplanned generator replacements
- Suitable for real-world deployment as a predictive maintenance decision-support tool

Final Model Performance on Test Set

Metric	Value
Accuracy	99.13%
Recall	93.17%
Precision	98.35%
F1 Score	95.59%

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Data Background and Contents

Data Background

- The dataset represents sensor readings collected from wind turbines operated by ReneWind
- Objective: To predict whether a wind turbine generator will fail based on sensor readings, so repairs can be done before a breakdown occurs
- This is a classification problem with:
  - ✓ Target variable: Target
  - ✓ Classes: Failure (1), No Failure (0)

Dataset Overview

- Contains only numerical features
- Data reflects:
  - ✓ Sensor readings from various turbine components
  - ✓ Environmental and mechanical measurements
- All feature values are continuous (float64)
- Features are ciphered — exact sensor type is confidential

Feature	Description
V1 – V10	Ciphered sensor reading - environmental or mechanical measurement from the turbine
V11 – V40	Ciphered sensor readings - same nature as above, covering additional turbine parameters
Target	Binary flag indicating generator status - 1 = Failure, 0 = No Failure

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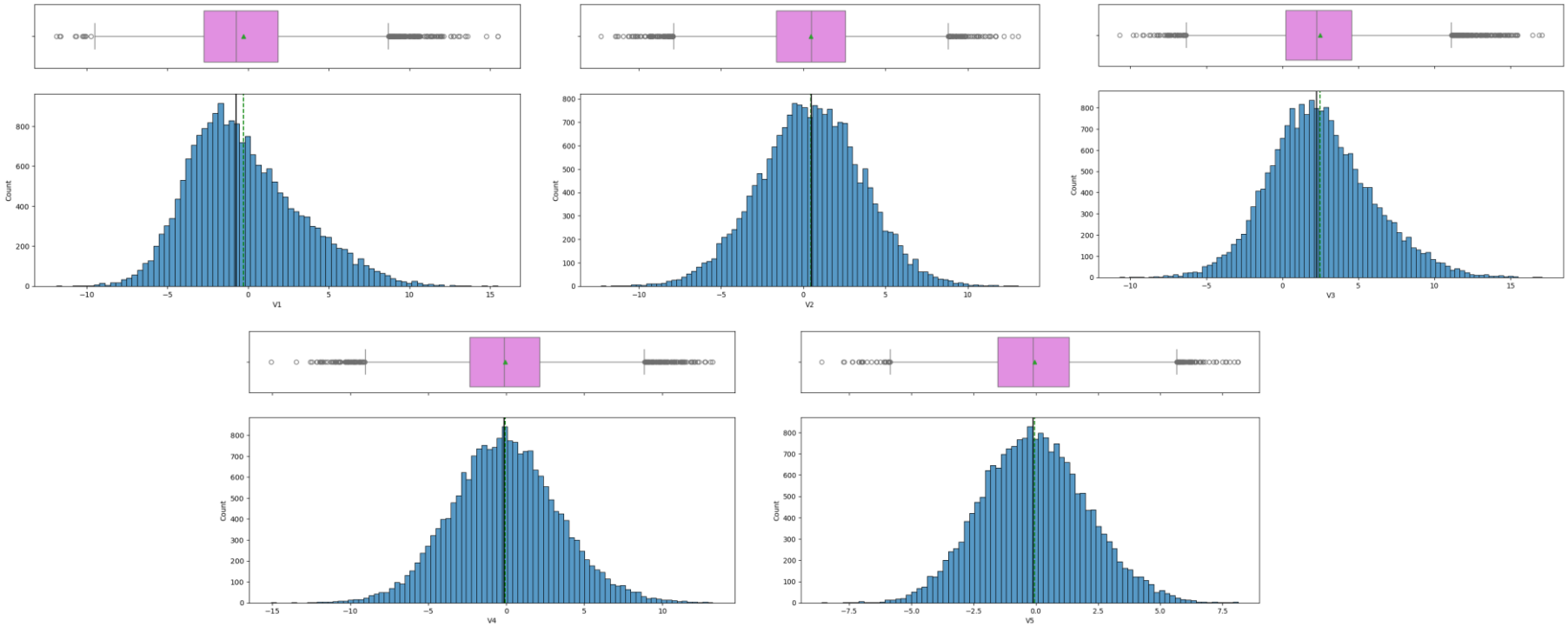
Univariate Analysis – Histogram & Box Plot (V1-V5)



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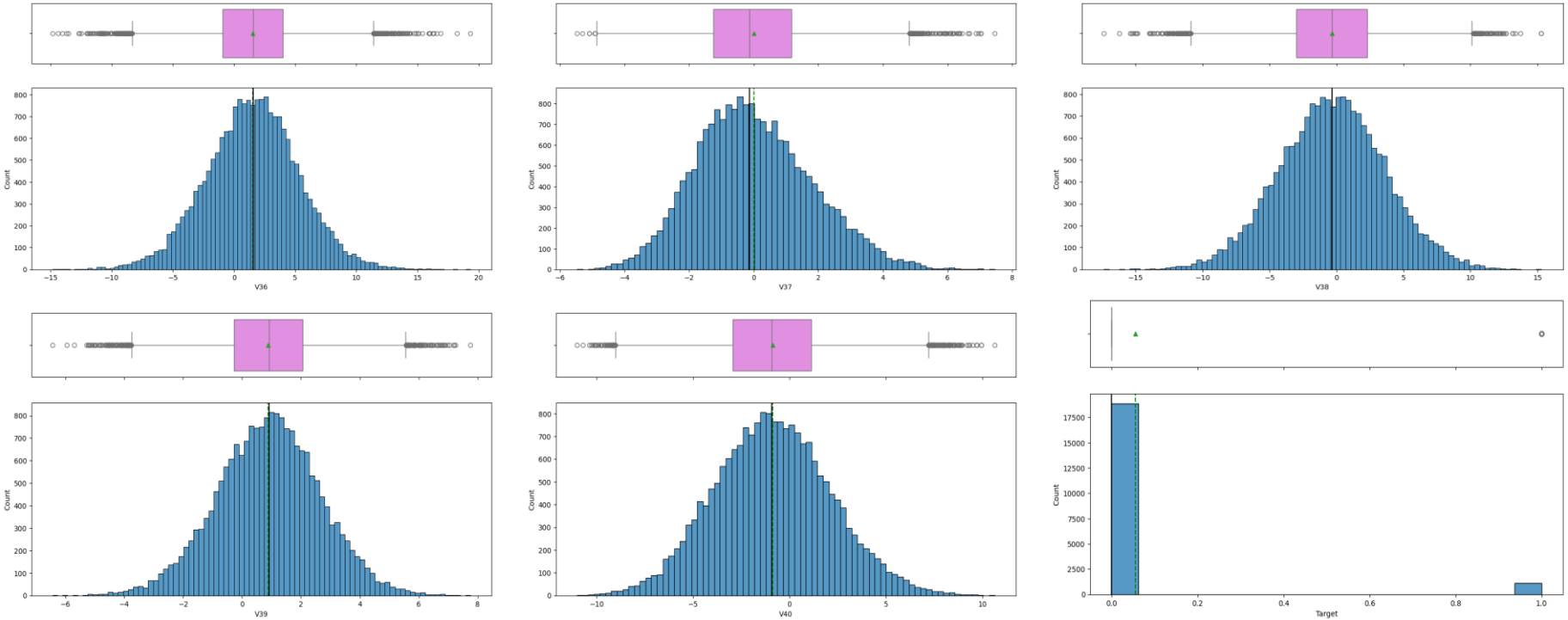
Univariate Analysis - Histogram & Box Plot (V36-V40 + Target)



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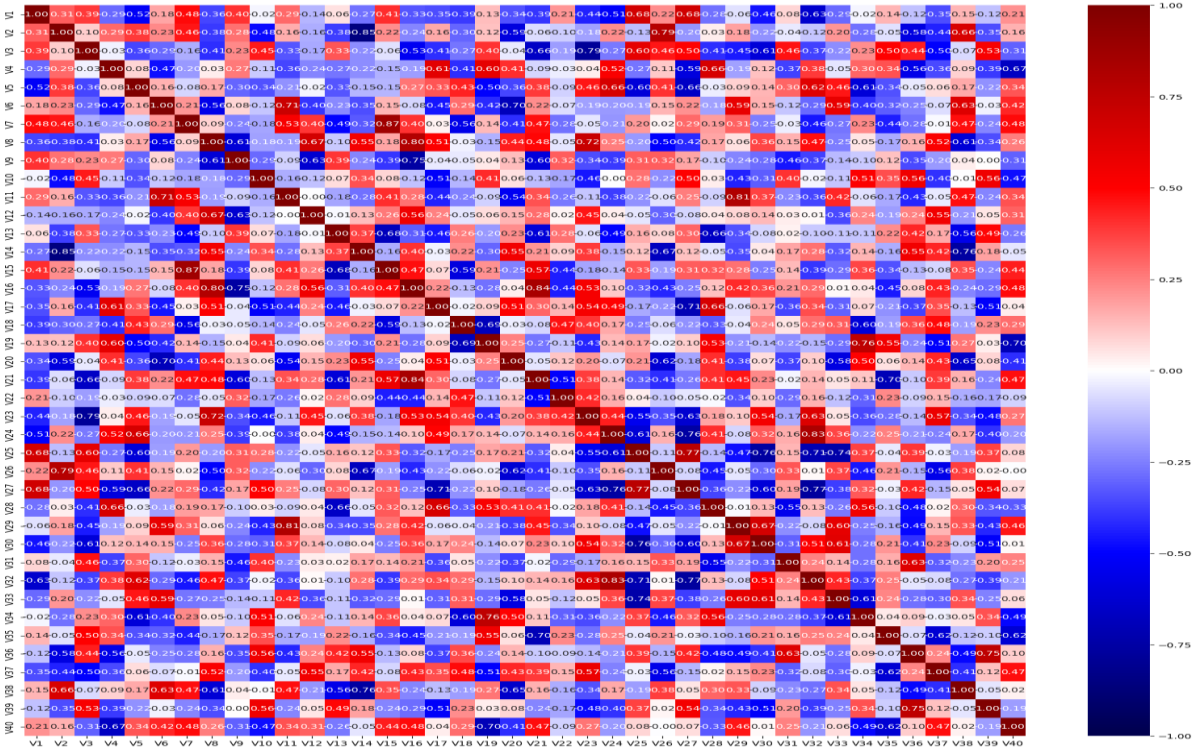
Bivariate Analysis – Correlation Heatmap



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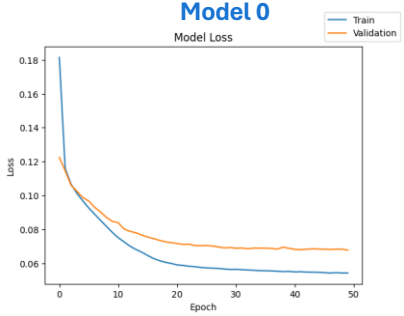
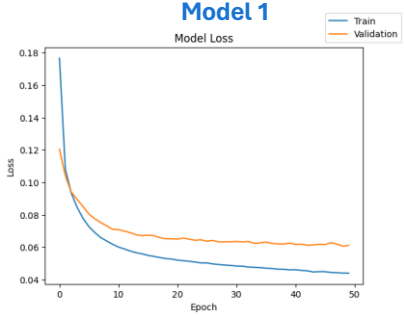
Model Building - Graphs

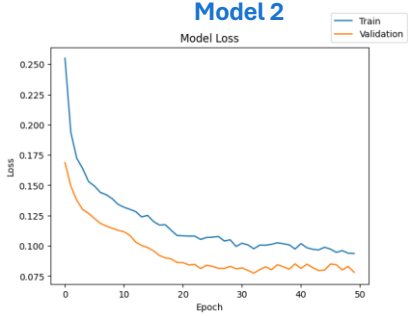
Model 0



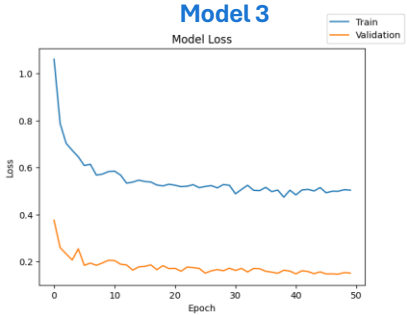
Model 1



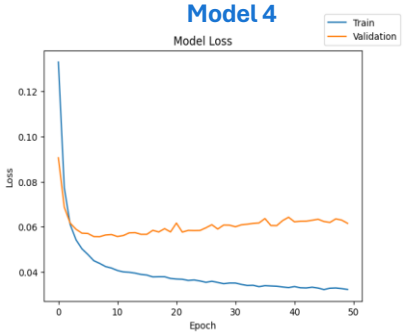
Model 2



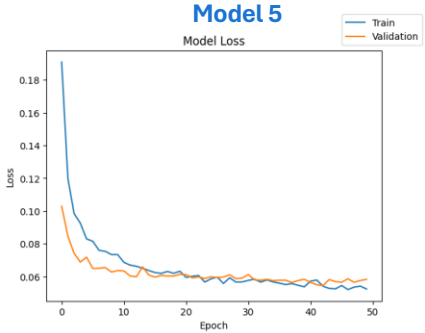
Model 3



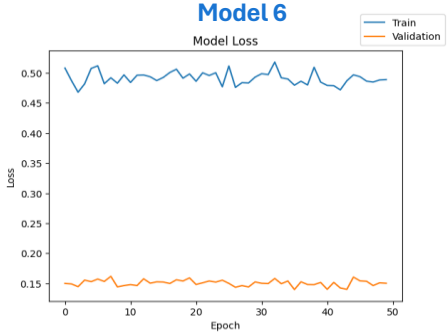
Model 4



Model 5



Model 6



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Univariate Charts

Additional univariate analysis visualizations were performed but are not included due to space constraints as they exceed over 30 charts

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